

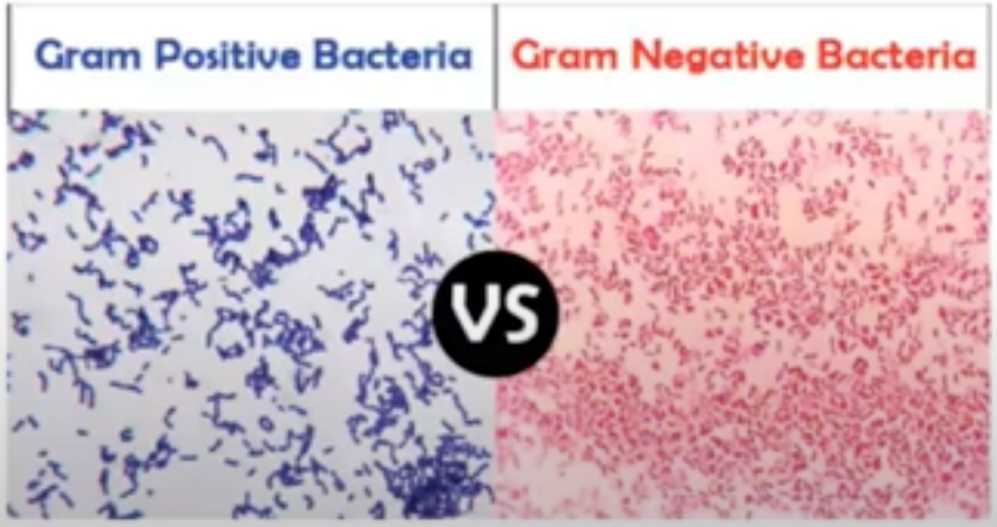
Principles of Antimicrobial Therapy

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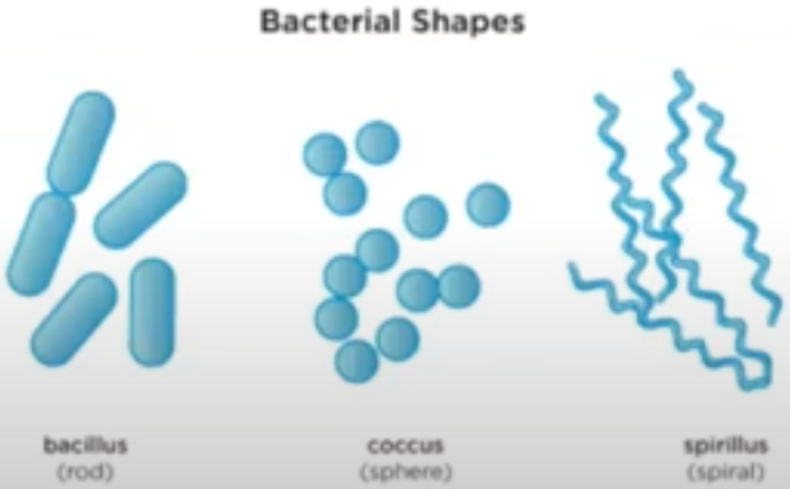
- **Antimicrobial drugs** are effective in the treatment of infections because of their **selective toxicity**; that is, they have the ability to injure or kill an invading microorganism without harming the host cells.

- In most instances, the selective toxicity is relative rather than absolute, requiring that the concentration of the drug be carefully controlled to attack the microorganism, while still being tolerated by the host.

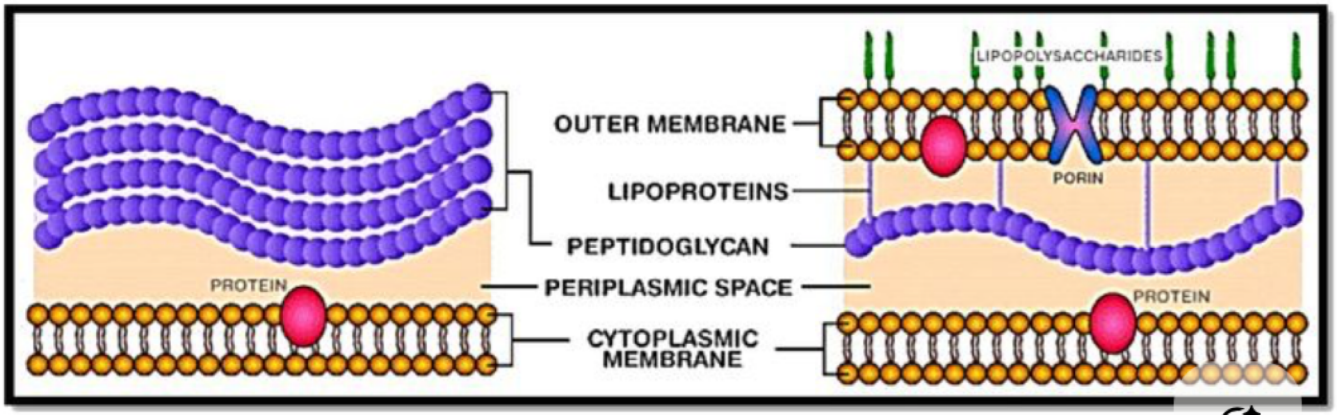
CLASSIFICATION OF MEDICALLY IMPORTANT BACTERIA



GRAM STAIN



SHAPES

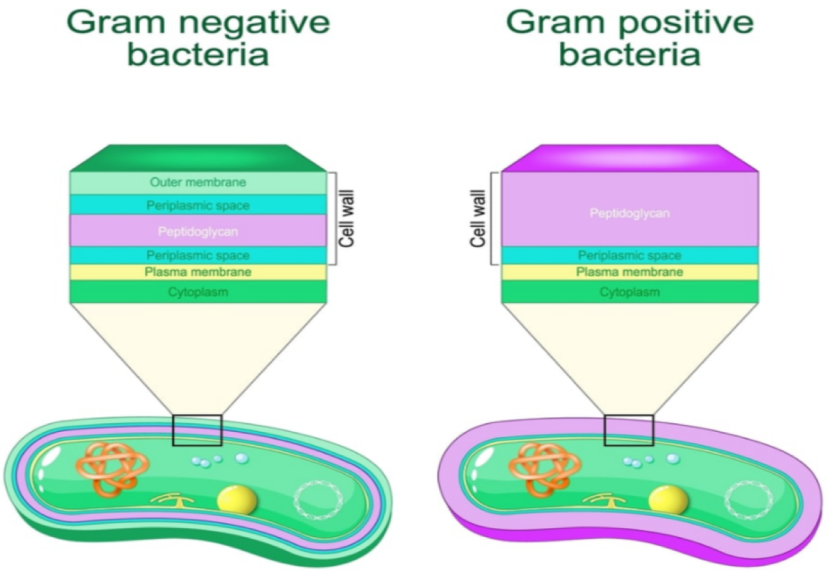


GRAM NEGATIVE BACTERIA

- **COCCI**
  - Nesseria
- **BACILLI/RODS**
  - Pseudomonas
  - Salmonella
  - Shigella
  - Klebsiella
  - Proteus
  - Escherichia
  - Yersinia
  - Bordetella
  - Haemophilus
  - Brucella
  - Pasteurella
  - Vibrio
  - Bacteriods
  - Fusobacterium
  - Prevotella
  - compylobacter
- **SPIROCHAETES**
  - Leptospira
  - Borrelia
  - Treponema

GRAM POSITIVE BACTERIA

- **COCCI**
  - Staphylococcus
  - Streptococcus
  - Enterococcus
  - Peptostreptococcus
- **BACILLI/ROD**
  - Corynebacterium
  - Listeria
  - Lactobacillus
  - Nocardia
  - Mycobacterium
  - Bacillus
  - Actinomyces
  - Clostridium



CLASSIFICATION OF ANTIMICROBIAL DRUGS

A. According to source:

- **Natural compounds:** e.g. penicillin, chloramphenicol.
- **Synthetic compounds:** e.g. sulfonamides, quinolones.
- **Semisynthetic compounds:** e.g. ampicillin.

- **B. According to the effect on microorganisms:**
- **Bactericidal agents:** that kills the microorganism e.g. penicillin.
- **Bacteriostatic agents:** arrest growth of the microorganism e.g. sulfonamides.
- **Note**: It is also possible for an antibiotic to be bacteriostatic for one organism and bactericidal for another. For example, **linezolid** is bacteriostatic against Staphylococcus aureus and enterococci, but is bactericidal against most strains of S. pneumoniae.

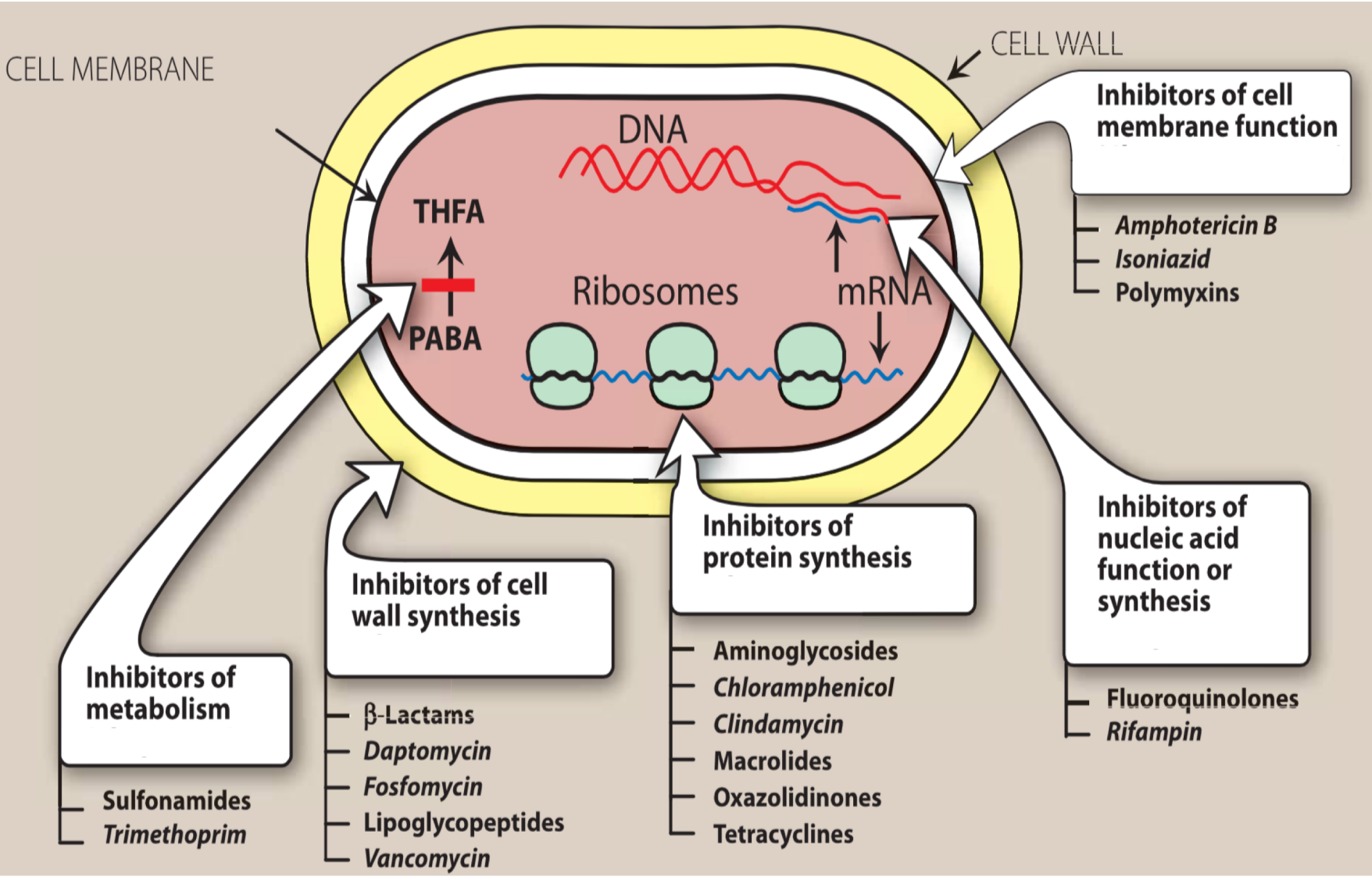
C. According to mechanism of action:

- Inhibition of cell wall synthesis: e.g. penicillin. Inhibition of cell membrane function e.g. amphotericin B and azoles.
- Agents act by inhibition of nucleic acid synthesis: e.g. quinolones.
- Agents act by inhibition of protein synthesis:

✓ Acting on ribosomal 30 S subunit e.g. aminoglycosides.

✓ Acting on ribosomal 50 S subunit e.g. macrolides.

- Inhibition of bacterial metabolism: e.g. sulfonamides.


CELL MEMBRANE
CELL WALL
DNA
THFA
PABA
Ribosomes
mRNA
Inhibitors of cell membrane function
Amphotericin B
Isoniazid
Polymyxins
Inhibitors of protein synthesis
Aminoglycosides
Chloramphenicol
Clindamycin
Macrolides
Oxazolidinones
Tetracyclines
Inhibitors of nucleic acid function or synthesis
Fluoroquinolones
Rifampin
Inhibitors of cell wall synthesis
β-Lactams
Daptomycin
Fosfomycin
Lipoglycopeptides
Vancomycin
Inhibitors of metabolism
Sulfonamides
Trimethoprim

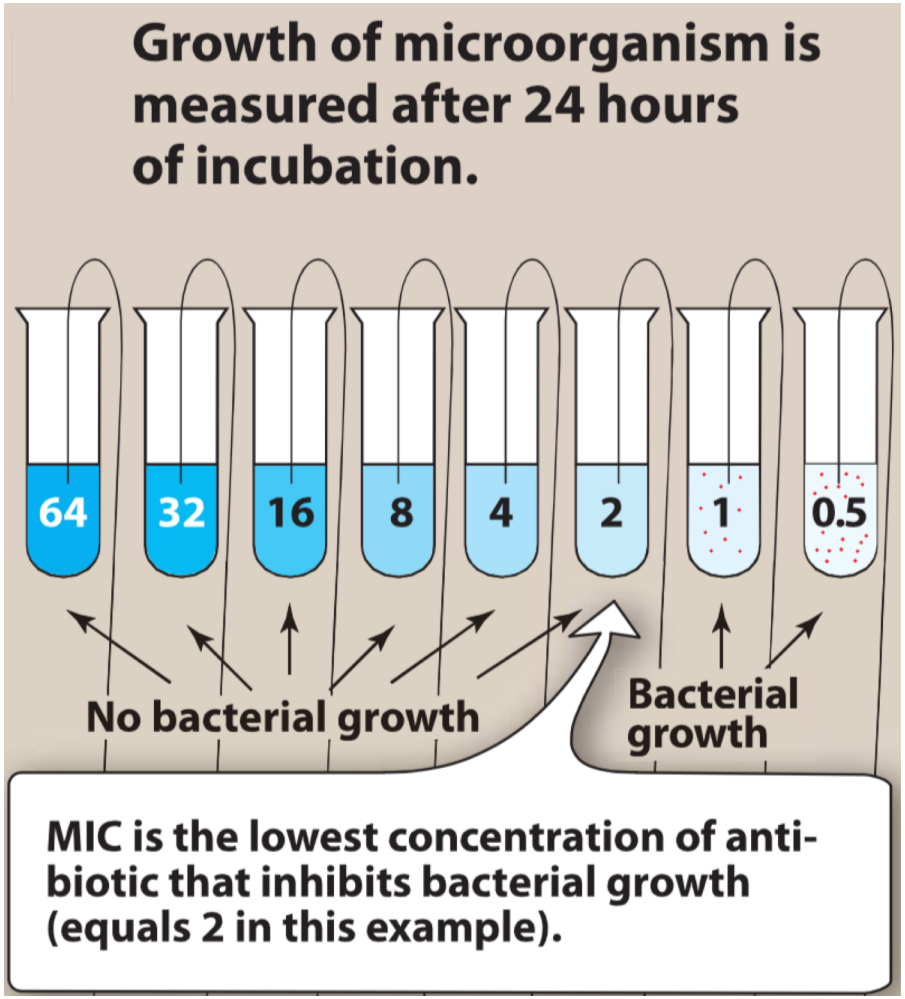
D. According to antimicrobial spectrum:

- Narrow-spectrum antibiotics: agents acting only on a single or a limited group of microorganisms eg: **isoniazid** is active only against Mycobacterium tuberculosis
- Extended-spectrum antibiotics: agents that are modified to be effective against gram-positive organisms and also against a significant number of gram-negative bacteria eg: **ampicillin**.
- Broad-spectrum antibiotics: agents affect a wide variety of microbial species. Drugs such as **tetracycline, carbapenems and fluoroquinolones.**

Determinants of the rational dosing

The **dose** and **frequency** of an antibiotic is based on three important properties. Understanding of these concepts can improve clinical outcomes and decrease the development of resistance.

Minimum inhibitory concentration (MIC): is the lowest antimicrobial concentration that prevents visible growth of an organism after 24 hours of incubation.



1. Concentration-dependent killing:

Certain antimicrobial agents, including **aminoglycosides** and **daptomycin**, show a significant increase in the rate of bacterial killing as the concentration of antibiotic increases from 4- to 64-fold the MIC of the drug for the infecting organism. Giving these antibiotics by a single large dose per day achieves high peak levels and cause rapid killing of bacteria.

2. Time-dependent killing:

β-lactam antibiotics, macrolides, clindamycin, and linezolid do not exhibit the concentration-dependent property; instead, the clinical efficacy depends on the time of the drug concentration remains above the MIC. So, preparations with long duration kill more bacteria.

3. Post-antibiotic effect (PAE):

the PAE is a persistent bacterial suppression after levels of antibiotic fall below the MIC. Antimicrobials with long PAE (e.g. aminoglycosides and fluoroquinolones) usually require one dose per day.

Combinations of antimicrobial drugs

- **Advantages of drug combinations**

- To obtain **synergism** Certain combinations of antibiotics, such as B-lactams and aminoglycosides.
- In **serious** bacterial infections e.g. bacterial meningitis or septicemia.
- To overcome **bacterial resistance** e.g. TB and pseudomonas infection.
- To **reduce toxicity** of one drug by using smaller doses of two drugs.

- **Disadvantages of drug combinations**

- Antagonism: when bactericidal drug acts by inhibiting cell wall synthesis (i.e. kills rapidly multiplying bacteria e.g. penicillin) is combined with bacteriostatic drug (e.g. erythromycin) which arrests the organism and prevents its multiplication

- development of antibiotic resistance by giving unnecessary combination therapy.

Drug resistance

The development of antibiotic resistance is a major problem in clinical practice.

- **Innate resistance:** It means natural resistance to drug due to micro-organism's basic structure or functional characteristics. For example,

✓ Gram -ve bactteria resistance to Vancomycin

✓ Gram +ve bactteria resistance to Aztreonam

✓ Anaerobes resistance to Aminoglycosides

✓ Mycoplasma species resistance to Penicillin & Cephalosporins

- **Acquired resistance:**

Acquired resistance is when bacteria that were sensitive to certain antibiotic become resistant with time. Mechanisms responsible for this resistance include:

1. Genetic alterations leading to drug resistance
2. Development of altered metabolic pathways.
3. Altered expression of proteins In drug-resistant organisms

- Production of enzymes that inactivate the drug.
- Alteration of drug binding site.

4. Decreased accumulation

Selection of antimicrobial agents

- **Identification of the infecting organism:**
  - Infected body tissue and secretion (e.g., blood, sputum, urine, wound drainage, etc.) must be sampled and cultured before initiating treatment.
  - **Empirical therapy:** before identification of the organism is necessary in critically ill patients and infections of unknown origin eg: meningitis, endocarditis, pyelonephritis.
  - **Prophylactic use of antibiotics:** certain clinical situations, such as dental procedures and surgeries, require the use of antibiotics for the prevention rather than for the treatment of infections (Pretreatment may prevent streptococcal infections in patients with a history of rheumatic heart disease, Pretreating of patients undergoing dental extractions who have implanted prosthetic devices, such as artificial heart valves).

- **Patient factors:**
- **In neonates**, the use of **chloramphenicol can lead to gray baby** syndrome because liver functions are not well developed. Also, the use of **sulfonamides may lead to kernicterus (**brain damage) due to displacement of bilirubin from serum albumin.
- **In growing children**, the use of **fluoroquinolones can lead to arthropathy** (joint swelling, stiffness and cartilage damage). Also, the use of **tetracyclines can bind to growing bones and teeth resulting in abnormal teeth and bone formation.**
- **In old age** (> 65 years) the incidence of renal toxicity with aminoglycosides is greater than in younger patients.
- **In immunocompromised patients,** the use of bactericidal agents is necessary, as the host's immune system is not capable of final elimination of the bacteria.
- **Pregnancy:** many antibiotics cross the placenta and cause adverse effects to the fetus e.g. **aminoglycosides and tetracyclines**.

- **Tissue penetration:**

✓ The capillary lining in some tissues, e.g. prostate, the vitreous body of the eye, and brain form natural barriers to drug delivery due to presence of tight junctions of th e capillary wall.

✓ Lipid soluble antibiotics e.g. chloramphenicol and metronidazole can cross these barriers in normal conditions. Penicillin is ionized at physiologic pH and cannot cross these barriers unless inflammation is present.

✓ Poor perfusion of some area, e.g. diabetic foot, reduces the amount of antibiotic reaching this area, making treatment is difficult.

Complications of antibiotic therapy

- **Hypersensitivity:** ex, penicillins, can cause serious hypersensitivity problems, ranging from urticaria (hives) to anaphylactic shock, rapid infusion of vancomycin causes "Red man syndrome".

- **Direct toxicity:** ex Aminoglycosides can cause ototoxicity, Chloramphenicol can cause bone marrow suppression. Fluoroquinolones can have effects on cartilage and tendons.

- **Superinfections:** Drug therapy, particularly with broad-spectrum antimicrobials or combinations of agents, can lead to alterations of the normal microbial flora.